Logistic Regression

Categorical response, and <u>cont</u> predictors.

Binary response (special case of categorical response).

$Y \in \{0, 1\}$, $X_1, \ldots, X_p$ <u>cont</u> predictors.

We want:

$$Y = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p + \epsilon \quad \text{where } \epsilon \sim N(0, \sigma^2)$$

$P(y) = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p$ model as a linear f of X.

$$\pi(\underline{x}) = P(Y = 1 \mid \underline{x} = x) = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p$$

LHS $\in [0, 1]$ RHS $\in \mathbb{R}$

Both sides are cont.

Take a f^n $g : [0, 1] \to \mathbb{R}$.

$$\text{Logit } g(y) = \ln\left(\frac{y}{1-y}\right)$$

Model:

$$\ln\left(\frac{\pi(\underline{x})}{1 - \pi(\underline{x})}\right) = \beta_0 + \beta_1 x_i + \cdots + \beta_p x_{\text{pi}} + \epsilon_i$$

$$\epsilon_i \sim N(0, \sigma^2)$$

g is a function that maps $[0, 1]$ to $\mathbb{R}$ and is monotonically increasing.

g is called a link function.

Examples:

$$g_1(y) = \ln\left(\frac{y}{1-y}\right) \text{ (logit link)}$$

$$g_2(y) = \mathbf{\Phi^{-1}(y)} \text{ (standard normal cdf)}$$

$\Phi(z) = P(Z \leq z)$ $\Phi : \mathbb{R} \to [0, 1]$

$Z \sim N(0, 1)$

$$\Phi^{-1} \text{ exists since } \Phi \text{ is cont and increasing.}$$

Φ^{-1} is increasing, $\Phi^{-1} : [0, 1] \to \mathbb{R}$.

$$\Phi^{-1}(\pi(\underline{x})) = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p + \epsilon$$

Probit link.

If X is cont r.v. with support $\mathbb{R}$, then the inverse cdf of X is a link f^n, e.g. t or double exponential.

Sometimes the physical problem indicates what is an appropriate model.

$g(y) \sim x$, to see if it is linear.

$$g(\pi(\underline{x}) = \beta_0 + \beta_1 x_1 \ldots$$

put $P(Y = 1 \mid x)$ in place of $\pi(\underline{x})$ Fit competing models for each, obtain residuals, carry out diagnostics & residuals are indep.

$g(y) = \ln\left(\frac{y}{1-y}\right)$

$$\begin{aligned}\frac{\pi}{1-\pi} &= \text{odds}, \quad \pi \text{ is a probability}\\ &= \frac{P(\text{success})}{P(\text{failure})}\end{aligned}$$

Model log odds as a linear f^n of the predictors $\rightarrow$ logistic regression.

$$\ln\left(\frac{\pi(\underline{x})}{1-\pi(\underline{x})}\right) = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p$$

β_0: log odds when all x_i are zeros.

When $p = 1$,

$$\begin{aligned}&\ln\left(\frac{\pi(x_1+1)}{1-\pi(x_1+1)}\right) - \ln\left(\frac{\pi(x_1)}{1-\pi(x_1)}\right)\\ &= \beta_0 + \beta_1(x_1+1) - \beta_0 - \beta_1 x_1 = \beta_1\end{aligned}$$

$$\text{log odds ratio} = \ln\left(\frac{\pi(x_1+1)}{1-\pi(x_1+1)} \Big/ \frac{\pi(x_1)}{1-\pi(x_1)}\right)$$

For general p:

$$\beta_i = \ln\left(\frac{\Pi(x_1, \ldots, x_i + 1, x_{i+1}, \ldots, x_p)}{1 - \Pi(x_1, \ldots, x_i + 1, x_{i+1}, \ldots, x_p)} \Big/ \frac{\Pi(x_1, \ldots, x_i, \ldots, x_p)}{1 - \Pi(x_1, \ldots, x_i, \ldots, x_p)}\right)$$

$$= (\beta_0 + \beta_1 x_1 + \cdots + \beta_i(x_i + 1) + \cdots + \beta_p x_p) - (\beta_0 + \cdots + \beta_i x_i + \cdots + \beta_p x_p)$$

β_i is log odds ratio when x_i is increased by 1 unit keeping all other x's fixed.

$$\pi(\underline{x}) = P(Y = 1 \mid X = \underline{x})$$

$$\ln\left(\frac{\Pi(\underline{x})}{1-\pi(\underline{x})}\right) = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p$$

Challenger Data

Space shuttle Challenger exploded right at the beginning of its flight on Jan 28, 1986. This was a serious disaster in the American space program.

The night before, an engineer had recommended to NASA that the shuttle should not be launched in such cold weather.

The temp was 31°F, the lowest temp of launch so far. The suggestion was overruled.

An inquiry commission was launched to see if there was evidence that predicted serious trouble related to low temperature.

There were no crashes before.

Failure of O-rings was a related event.

These are rubber rings that fill the gaps between parts of the rocket.

If there is a leak, hot gas pushes out, catches fire, and burns the rocket.

These instances had happened in past.

$$x = \text{Temperature (launching temp)}$$
$$y = \text{Failure of O-Rings}$$

$x = 31, \quad y = ? \quad P(Y = 1)$